

Incr., Decr., Local Extrema

Section 3.3

9/24/25

A: What we ^{should} know:

- 1) $f(x)$ is increasing when $f'(x) > 0$
- 2) $f(x)$ is decreasing when $f'(x) < 0$

LOCAL Extrema: 1st Derivative Test

1) $f(x)$ has a local max when $f'(x)$ changes from positive to negative

2) $f(x)$ has a local min when $f'(x)$ changes from negative to positive.

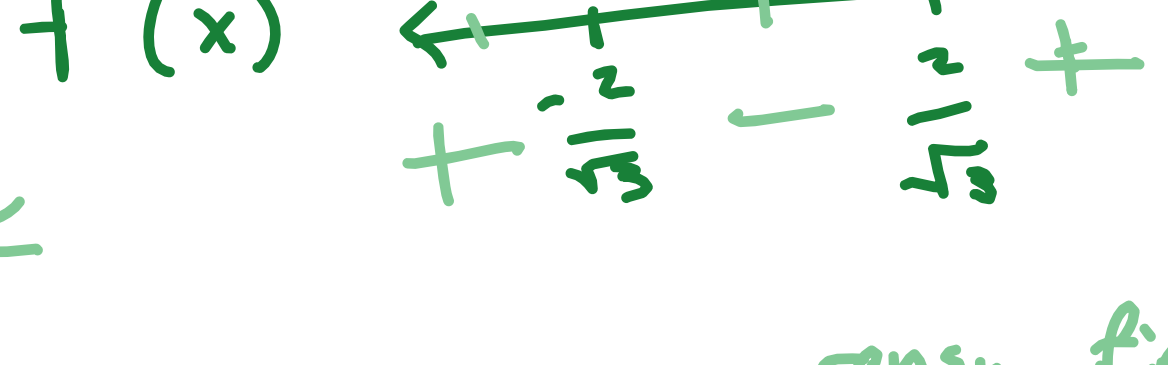
Ex1:

$$f(x) = x^3 - 4x$$

@w incr./decr
local max/min

$$f'(x) = 3x^2 - 4 = 0$$

$$x = \pm \frac{2}{\sqrt{3}}$$



plug in between ranges, find our signs

$f(x)$ is increasing: $(-\infty, -\frac{2}{\sqrt{3}})$
 $(\frac{2}{\sqrt{3}}, \infty)$

because $f'(x) > 0$

$f(x)$ is decreasing: $(-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}})$

because $f'(x) < 0$

$f(x)$ is local max: $(-\frac{2}{\sqrt{3}}, (-\frac{2}{\sqrt{3}})^3 - 4(-\frac{2}{\sqrt{3}}))$

because $f'(x)$ changes from Positive to Negative

Ex2: $x\sqrt{4-x}$

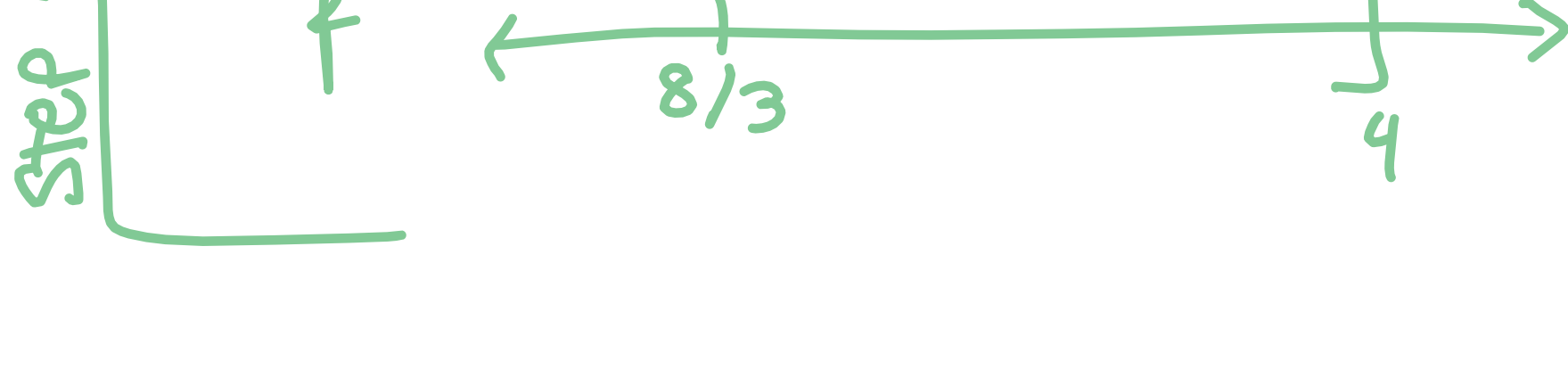
$$f'(x) = \sqrt{4-x} + \frac{1}{2}x(4-x)^{-1/2}(-1) = 0$$

$$\sqrt{4-x} - \frac{x}{2\sqrt{4-x}} = 0$$

$$\frac{2(4-x) - x}{2\sqrt{4-x}} = \frac{8-3x}{2\sqrt{4-x}} = 0$$

$$x = 8/3$$

$$x = 4$$



f incr. : $(-\infty, 8/3)$ because $f'(x) > 0$

f decr: $(8/3, 4]$ because $f'(x) < 0$

f local max: @ $x = 8/3$, $f'(x)$ goes from $+$ to $-$

f local min: N/A

EX3: $f(x) = \frac{x^2}{2x-1}$

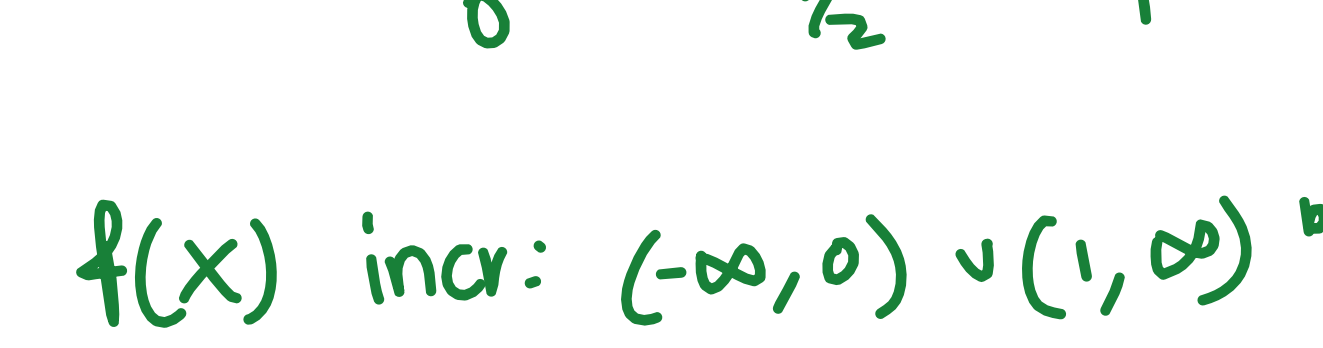
$$f'(x) = \frac{2x(2x-1) - x^2(2)}{(2x-1)^2}$$

$$\frac{-2x + 4x^2 - 2x^2}{(2x-1)^2}$$

$$\frac{2x^2 - 2x}{(2x-1)^2} = 0$$

$$\frac{2x(x-1)}{(2x-1)^2} = 0$$

$$x = 0, x = 1, x = 1/2$$



$f(x)$ incr: $(-\infty, 0) \cup (1, \infty)$ b/c $f'(x) > 0$

$f(x)$ decr: $(0, 1/2) \cup (1/2, 1)$ b/c $f'(x) < 0$

$f(x)$ local max: $(0, 0)$ $f'(x)$ goes from $+$ to $-$

$f(x)$ local min: $(1, 1)$ $f'(x)$ goes from $-$ to $+$